

Primal–Dual Relationships in Convex Optimization

Motivation Questions

Before we study primal–dual optimization, let us think about the following questions:

- ▶ What is a **primal problem**?
- ▶ Is the concept of a primal problem used only in **optimization**?
- ▶ If we already have the primal problem, **why do we introduce the dual problem**?
- ▶ What difficulties arise when solving the primal problem directly?
- ▶ What exactly is the **dual problem**?
- ▶ Why do many optimization methods focus on the **dual formulation** instead of the primal?

Goal of this lecture:

- ▶ Understand the relationship between **primal and dual problems**.
- ▶ Learn how to derive the **dual problem using the Lagrangian**.
- ▶ See examples where solving the **dual problem is easier**.

General Convex Optimization Problem

Primal problem:

$$\begin{array}{ll}\min_x & f(x) \\ \text{s.t.} & g_i(x) \leq 0 \quad i = 1, \dots, m\end{array}$$

Assumptions:

- ▶ $f(x)$ convex
- ▶ $g_i(x)$ convex

Feasible set:

$$\{x \mid g_i(x) \leq 0, \}$$

Lagrangian Function

To handle constraints we define the Lagrangian:

$$L(x, \lambda) = f(x) + \sum_{i=1}^m \lambda_i g_i(x)$$

where

$$\lambda_i \geq 0$$

Interpretation:

- ▶ λ_i penalizes violation of inequality constraints.
- ▶ Converts constrained optimization into unconstrained optimization.

Dual Function

Dual function definition:

$$g(\lambda) = \inf_x L(x, \lambda)$$

Properties:

- ▶ $g(\lambda)$ is always concave.
- ▶ Provides a lower bound on the primal optimal value.

Weak duality:

$$g(\lambda) \leq p^*$$

where p^* is the optimal value of the primal problem.

Dual Problem

The dual problem is defined as

$$\max_{\lambda} g(\lambda)$$

subject to

$$\lambda \geq 0$$

Let

$$d^* = \max g(\lambda)$$

Relationship between primal and dual:

$$d^* \leq p^*$$

The dual searches for the tightest lower bound on the primal objective.

Strong Duality

Strong duality holds when

$$p^* = d^*$$

A sufficient condition is **Slater's condition**.

Slater condition:

There exists x such that

$$g_i(x) < 0$$

for all inequality constraints.

Implication:

- ▶ Dual optimum equals primal optimum.
- ▶ Dual variables provide information about the primal solution.

Example 1: Simple Intuition

Primal problem:

$$\min_x x^2$$

subject to

$$x \geq 1$$

Optimal solution:

$$x^* = 1$$

Optimal value:

$$p^* = 1$$

Dual Derivation

Rewrite constraint:

$$1 - x \leq 0$$

Lagrangian:

$$L(x, \lambda) = x^2 + \lambda(1 - x)$$

Stationarity condition:

$$\frac{\partial L}{\partial x} = 2x - \lambda = 0$$

$$x = \frac{\lambda}{2}$$

Dual function:

$$g(\lambda) = -\frac{\lambda^2}{4} + \lambda$$

Dual problem:

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Example 2: Equality Constrained Quadratic

Primal problem:

$$\min_x \frac{1}{2} x^T Q x + c^T x$$

subject to

$$Ax = b$$

Lagrangian:

$$L(x, \lambda) = \frac{1}{2} x^T Q x + c^T x + \lambda^T (Ax - b)$$

Stationarity condition:

$$Qx + c + A^T \lambda = 0$$

$$x = -Q^{-1}(c + A^T \lambda)$$

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Dual Problem

Dual function becomes

$$g(\lambda) = -\frac{1}{2}(c + A^T \lambda)^T Q^{-1}(c + A^T \lambda) - b^T \lambda$$

Dual optimization:

$$\max_{\lambda} g(\lambda)$$

Example 3: Linear Programming

Primal problem:

$$\min c^T x$$

subject to

$$Ax \leq b$$

Lagrangian:

$$L(x, \lambda) = c^T x + \lambda^T (Ax - b)$$

$$= (c + A^T \lambda)^T x - b^T \lambda$$

Dual constraint:

$$A^T \lambda + c = 0$$

Dual problem:

$$\max -b^T \lambda$$

Example 4: Support Vector Machine

Primal problem:

$$\min_{w,b} \frac{1}{2} \|w\|^2$$

subject to

$$y_i(w^T x_i + b) \geq 1$$

Dual formulation:

$$\max_{\alpha} \sum \alpha_i - \frac{1}{2} \sum_{i,j} \alpha_i \alpha_j y_i y_j x_i^T x_j$$

subject to

$$\alpha_i \geq 0$$

$$\sum \alpha_i y_i = 0$$

Classroom Exercise 1

Primal problem:

$$\min_x \frac{1}{2} ||x||^2$$

subject to

$$Ax = b$$

Tasks:

- ▶ Form the Lagrangian
- ▶ Compute stationarity condition
- ▶ Derive the dual function
- ▶ Write the dual problem

Classroom Exercise 2

Primal problem:

$$\min_x \sum x_i \log x_i$$

subject to

$$\sum x_i = 1$$

Tasks:

- ▶ Write the Lagrangian
- ▶ Derive stationary condition
- ▶ Compute optimal x_i

Result:

$$x_i = e^{-1-\lambda}$$

Key Takeaways

- ▶ Every convex optimization problem has a dual formulation.
- ▶ Dual provides lower bounds on the primal objective.
- ▶ Under strong duality

$$p^* = d^*$$

- ▶ Duality is widely used in machine learning, economics, and statistics.